[548] (CONTINUED)

On Listing's Theorem.

[From the Messenger of Mathematics, vol. I. (1872), pp. 81–88 — continuation.]

b. It might be possible to find an analogous definition, but the most simple one is that a denotes the number of ovoids (unilinked ovoids) or other surfaces not bounded by any point or line.

d is the number of spaces, reckoning as one of them infinite space.

a'' is the sum, for the several spaces, of the number of circuits in each space: the word circuit here signifying a path in the space from a point to itself; all circuits which can by continuous variation be made to coincide being considered as identical, and the connectible circuit reducible to the point itself being throughout disregarded. Moreover, we count only the simple circuits, disregarding any circuit which can be obtained by a repetition or combination of these. Thus for infinite space, or for the space within an ovoid, there is only the evanescent circuit, or there is no circuit to be counted; and the same is the case if within such space we have any number of ovoidal blocks (the term will, I think, be understood without explanation); but if within the space we have an oval, ring, or other ring-block of any kind whatever, then there is therefore the connectible circuit, a circuit interlacing the ring-block, and we count one circuit; and so if there are a ring-blocks, either separate or interlacing each other in any manner, then there are, and we accordingly count, a circuits. So for the space inside a ring there is (besides the connectible circuit), and we count, one circuit; and the case is the same if we have within the ring any number of ovoidal blocks whatever; but if there is within the ring an oval ring or other ring-block, then there is one new circuit, and we count in all (for the space in question) two circuits.

p is the number of detached parts of the figure; or, say the number of detached aggregates of points, lines, and surfaces. Observe, that rings interlacing each other in any manner (but not intersecting) are considered as detached; as also two closed surfaces, one within the other, are considered as detached. The figure may be infinite space alone; we have then $p = 0$.

The complete which follow will further illustrate the meaning of the terms and nature of the theorem; and will also indicate in what manner a general demonstration of the theorem might be arrived at.

1. Infinite space.

$$\begin{array}{llllll}
a=0, & A=0, & & & & \\
b=0, & a'=0, & B=0, & & & \\
c=0, & a''=0, & w=0, & C=0, & & \\
d=1, & a'''=0, & & D & & =1 \\
p=0, & & & p-1 & & =-1 \\
 & & & \overline{0} & & =\overline{0}.
\end{array}$$

2. Spherical surface.

$$\begin{array}{llllll}
a=0, & A=0, & & & & \\
b=0, & a'=0, & B=0, & & & \\
c=1, & a''=0, & w=1, & C=1, & & \\
d=2, & a'''=0, & & D & & =2 \\
p=1, & & & p-1 & & =0 \\
 & & & \overline{2} & & =\overline{2}.
\end{array}$$

viz. the effect is to increase C by 2 and D and $p-1$ each by 1.

3. Spherical surface, with point upon it.

$$\begin{array}{llllll}
a=1, & A=1, & & & & \\
b=0, & a'=0, & B=0, & & & \\
c=1, & a''=0, & w=1, & C=1, & & \\
d=2, & a'''=0, & & D & & =2 \\
p=1, & & & p-1 & & =0 \\
 & & & \overline{2} & & =\overline{2}.
\end{array}$$

viz. the effect is to increase a and diminish c each by 1: that is, A is increased and C diminished each by 1.

4. Spherical surface with two points.

$$\begin{array}{llllll}
a=2, & A=2, & & & & \\
b=0, & a'=0, & B=0, & & & \\
c=1, & a''=0, & w=1, & C=1, & & \\
d=2, & a'''=0, & & D & & =2 \\
p=1, & & & p-1 & & =0 \\
 & & & \overline{2} & & =\overline{2}.
\end{array}$$

viz. the second point increases a and a' each by 1, that is, it increases A and diminishes C each by 1.

And for each new point on the spherical surface there is this same effect; we may have, for the next case:

5. Spherical surface with points $(n \geq 2)$.

$$\begin{array}{llllll}
a=n, & A=n, & & & & \\
b=0, & a'=0, & B=0, & & & \\
c=1, & a''=n-1, & w=0, & C=2-n, & & \\
d=2, & a'''=0, & & D & & =2 \\
p=1, & & & p-1 & & =0 \\
 & & & \overline{2} & & =\overline{2}.
\end{array}$$

Imagine that besides the n points there is an aperture (bounded by a closed curve); the case is:

6. Spherical surface with n points ($n \geq 2$) and aperture.

$$\begin{array}{llllll}
a = n, & A = n, & & & & \\
b = 1, & a' = 1, & B = 0, & & & \\
c = 1, & a'' = n, & w = 0, & C = 1 - n, & & \\
d = 1, & a''' = 0, & & D & & = 1 \\
p = 1, & & & p - 1 & & = 0 \\
 & & & \overline{1} & & = \overline{1}.
\end{array}$$

viz. b and a' are each increased by 1, and therefore B is still unaltered; a'' is increased, and therefore C diminished, by 1; a''' is increased, and therefore D diminished, by 1, and each new aperture produces the like effect. Thus we have:

7. Spherical surface with apertures ($n \geq 2$) and two apertures.

$$\begin{array}{llllll}
a = n, & A = n, & & & & \\
b = 2, & a' = 2, & B = 0, & & & \\
c = 1, & a'' = n + 1, & w = 0, & C = -n, & & \\
d = 0, & a''' = 0, & & D & & = 0 \\
p = 1, & & & p - 1 & & = 0 \\
 & & & \overline{0} & & = \overline{0}.
\end{array}$$

viz. b and a' are each increased by 1, and therefore B is still unaltered; a'' is increased, and therefore C diminished, by 1; a''' is increased, and therefore D diminished by 1, and each new aperture produces the like effect. Thus we have:

8. Spherical surface with apertures ($n \geq 2$).

$$\begin{array}{llllll}
a = 0, & A = 0, & & & & \\
b = n, & a' = n, & B = 0, & & & \\
c = 1, & a'' = n - 1, & w = 0, & C = 2 - n, & & \\
d = 1, & a''' = n - 1, & & D & & = 2 - n \\
p = 1, & & & p - 1 & & = 0 \\
 & & & \overline{2 - n} & & = \overline{2 - n}.
\end{array}$$

where, comparing with case 5, we see the different effects of a point and an aperture.

9. Spherical surface with n apertures ($n \geq 2$) and a point or points on each or any of the bounding curves of the apertures.

   If on the bounding curve of any aperture we place a point, this increases a, and therefore A, by 1; the bounding curve is no longer a simple closed curve, and we thus also have a diminished, and therefore B increased by 1; and the balance still holds.

Placing on the same bounding curve a second point, a, and therefore A, is increased by 1, but the bounding curve is converted into two distinct curves, that is, b, and therefore A, is increased by 1, and the balance still holds. And the like for each new point on the same bounding curve.

10. Spherical surface with a point connected in any manner by lines.

    Reverting to the cases 4 and 5, by joining any two points by a line, we increase b and therefore B by 1; but as regards a' the two united points take effect as a single point; that is, a' is diminished and therefore C increased, by 1, the balance is therefore unbalanced.

    The case is the same for each new line, if only we do not thereby produce on the surface a closed polygon, or partition or existing closed polygon: in each of these cases we also increase b and therefore B by 1; and instead of diminishing a', we increase c by 1, and therefore still increase C by 1, and the balance continues to subsist.

    By continuing to join the several points we at last arrive at a spherical surface partitioned into polygons in any manner whatever; or, what is the same thing, we have

11. Closed polyhedral surface. Here, if S is the number of summits, F the number of faces, E the number of edges; then

$$\begin{array}{lllll} a = S, & A = S, & & & \\ b = E, & a' = 0, & B = E & & \\ c = F, & a'' = 0, & w = 0, & C = -F, & \\ d = 2, & a''' = 0, & & D & = 2 \\ p = 1, & & & p - 1 & = 0 \\ & & & \overline{S + F} & = \overline{E + 2}, \end{array}$$

as that we have Euler's theorem. Observe that this theorem (Euler's) does not apply to annular polyhedral surface, or to polyhedral shells. For instance, consider a shell, the exterior and interior surfaces of which are each of them a closed polyhedral surface; $S = S' + S''$, $F = F' + F''$, $E = E' + E''$, where $S' + F' = E' + 2$, $S'' + F'' = E'' + 2$, and therefore $S + F = E + 4$. Listing's theorem, of course, applies, viz. we have

12.

$$\begin{array}{lll} a = S' + S'', & A = S' + S'', & \\ b = E' + E'', & B = E' + E'' & \\ c = F' + F'', & C = -F' - F'', & \\ d = 3, & D & = 3 \\ p = 1, & p - 1 & = 0 \\ & \overline{S + F} & = \overline{E + 3}. \end{array}$$

As another group of examples, consider a plane rectangle, for instance, a sheet of paper bounded by its four edges; here

13.

$$\begin{array}{lllll} a=4, & A=4, & & & \\ b=4, & a'=0, & B=4 & & \\ c=1, & a''=0, & w=0, & C=-1, & \\ d=1, & a'''=0, & & \quad D & =1 \\ p=1, & & & \quad p-1 & =0 \\ & & & \quad \overline{5} & =\overline{5}. \end{array}$$

Let the paper be formed into a tube by uniting two opposite sides, the suture not being obliterated, but continuing as a line drawn lengthwise from one extremity of the tube to the other; here

14.

$$\begin{array}{lllll} a=2, & A=2, & & & \\ b=3, & a'=0, & B=3 & & \\ c=1, & a''=0, & w=0, & C=-1, & \\ d=1, & a'''=0, & & \quad D & =1 \\ p=1, & & & \quad p-1 & =0 \\ & & & \quad \overline{3} & =\overline{3}. \end{array}$$

Let the suture be obliterated, or that we have simply a tube open at each end; here

15.

$$\begin{array}{lllll} a=0, & A=0, & & & \\ b=2, & a'=0, & B=2 & & \\ c=1, & a''=0, & w=0, & C=-1, & \\ d=1, & a'''=0, & & \quad D & =1 \\ p=1, & & & \quad p-1 & =0 \\ & & & \quad \overline{1} & =\overline{1}. \end{array}$$

Let the tube be formed into an annulus by bending it round and joining the two extremities, the suture not being obliterated but continuing as the closed curve round the tube; here

16.

$$\begin{array}{lllll} a=0, & A=0, & & & \\ b=1, & a'=0, & B=1 & & \\ c=1, & a''=0, & w=0, & C=-1, & \\ d=1, & a'''=0, & & \quad D & =0 \\ p=1, & & & \quad p-1 & =0 \\ & & & \quad \overline{0} & =\overline{0}. \end{array}$$

Let the suture be obliterated, so that we have simply a tubular annulus; here

17.

$$\begin{array}{llll}
a=0, & A=0, & & \\
b=0, & a'=0, & B=0 & \\
c=1, & a''=1, & w=0, & C=0, \\
d=2, & a'''=0, & & \quad D \qquad\qquad =0 \\
p=1, & & & \quad p-1 \qquad =0 \\
 & & & \quad \overline{0} \qquad\qquad =\overline{0}.
\end{array}$$

We may compare herewith the case of a simple annulus or closed curve.

18.

$$\begin{array}{llll}
a=0, & A=0, & & \\
b=1, & a'=1, & B=0 & \\
c=0, & a''=0, & w=0, & C=0, \\
d=1, & a'''=0, & & \quad D \qquad\qquad =0 \\
p=1, & & & \quad p-1 \qquad =0 \\
 & & & \quad \overline{0} \qquad\qquad =\overline{0}.
\end{array}$$

Add to such an annulus, for instance, three radii meeting in the centre; thus

19.

$$\begin{array}{llll}
a=4, & A=4, & & \\
b=6, & a'=0, & B=6 & \\
c=0, & a''=0, & w=0, & C=0, \\
d=1, & a'''=3, & & \quad D \qquad\qquad =-2 \\
p=1, & & & \quad p-1 \qquad =0 \\
 & & & \quad \overline{-2} \qquad\quad =\overline{-2}.
\end{array}$$

Let the last-mentioned figure become tubular, all sutures being obliterated; then

20.

$$\begin{array}{llll}
a=0, & A=0, & & \\
b=0, & a'=0, & B=0 & \\
c=1, & a''=4, & w=1, & C=-4, \\
d=2, & a'''=0, & & \quad D \qquad\qquad =2 \\
p=1, & & & \quad p-1 \qquad =0 \\
 & & & \quad \overline{-4} \qquad\quad =\overline{-4}.
\end{array}$$

And so if instead of the tubular figure, annulus with three radii, we had a tubular figure, annulus with diameter, then

21.

$$\begin{array}{lllll}
a = 0, & A = 0, & & & \\
b = 0, & a' = 0, & B = 0 & & \\
c = 1, & a'' = 2, & w = 1, & C = -2, & \\
d = 2, & a''' = 0, & & D & = 2 \\
p = 1, & & & p - 1 & = 0 \\
 & & & \overline{-2} & = \overline{-2}.
\end{array}$$

and the like in other cases.

[549]

Note on the Maxima of Certain Factorial Functions.

[From the Messenger of Mathematics, vol. II. (1873), pp. 129, 130.]

I CONSIDER the functions

$$\begin{aligned}
y_1 &= x(x-1),\\
y_2 &= x(x-\tfrac{1}{2})(x-1),\\
y_3 &= x(x-\tfrac{1}{3})(x-\tfrac{2}{3})(x-1),\\
&\vdots\\
y_p &= x\left(x-\tfrac{1}{p}\right)\left(x-\tfrac{2}{p}\right)\cdots\left(x-\tfrac{p-1}{p}\right)(x-1).
\end{aligned}$$

Attending only to the absolute values, disregarding the signs, y_p has a maximum, viz. if x be odd, $= \frac{1}{2}p+1$ suppose, these are

$$Y_1,\ Y_3,\ \ldots,\ Y_p,\ Y_{p+2},\ Y_{p+4},\ \ldots,\ Y_n,$$

where Y_{p+m} corresponds to the value $x = \frac{1}{2p} + \frac{m}{p}$, viz. to values of x between

$$0 \text{ and } \tfrac{1}{2p},\ \tfrac{1}{p} \text{ and } \tfrac{2}{p},\ \ldots,\ \tfrac{p-1}{p} \text{ and } \tfrac{p+1}{2p},\ \text{ and } \tfrac{1}{2}.$$

But if n be even, $= 2p$ suppose, these the maxima are

$$Y_1,\ Y_3,\ \ldots,\ Y_{p-1},\ Y_p,\ Y_{p+2},\ \ldots,\ Y_n,$$

where Y_{p+m} corresponds to values of x between

$$0 \text{ and } \tfrac{1}{2p},\ \tfrac{1}{p} \text{ and } \tfrac{2}{p},\ \ldots,\ \tfrac{p-1}{p} \text{ and } \tfrac{1}{2}.$$

In every case the maximum decreases from Y_1, which is the greatest, to Y_p or Y_{p+1}, which is the least; in particular, $n = 2p+1$, there

$$Y_{p+r} = \left[\left(\tfrac{r}{2p+1}\right)\cdots\left(\tfrac{1}{r}\right)\right](-1)$$

$$= \left(\tfrac{2p-1}{1}\right)\tfrac{r-1}{p+1}\cdots\tfrac{p-r-2}{p+1}$$

$$= \pm\left[\frac{1\cdot 3\cdots(2p-1)}{2^{p+1}p^{p+1}}\right]^{1/r}\frac{(2p-r+1)}{2p+1}\left(\frac{1}{r}\right)\cdots\left(\frac{1}{p}\right)\frac{(r-1)!}{(p+r)!}$$

which is

$$= \pm\frac{\Gamma\left(\tfrac{1}{2}+\tfrac{r}{2p+1}\right)\Gamma\left(\tfrac{1}{2}-\tfrac{r}{2p+1}\right)}{(2p+1)^{2p+1}}\cdot\frac{\Gamma(p+1-r)}{\Gamma(p+1)\Gamma(p+r+1)}.$$

Suppose p is large; then, as for large values of x,

$$\Gamma(x+a) = \sqrt{2\pi}\,x^{x+a-\frac{1}{2}}e^{-x},$$

we have

$$\Gamma\left(p+\tfrac{1}{2}\right) = \sqrt{2\pi}(p)^p e^{-p} + \frac{1}{12p} + \cdots$$

$$(2p+1)^{2p+1} = (2p)^{2p}\left(1+\tfrac{1}{2p}\right)^{2p}\left(1+\tfrac{1}{2p}\right) = \frac{e^{2p}}{2p}\,p^p\,e^{-p},$$

and as

$$Y_{p+r} = \frac{2^{2p}p^{2p-1}}{e^{2p}}\,\frac{e^{2p}}{2p}\,p^p e^{-p}\left(\frac{1}{2}\right)^{p+1}.$$

Also Y_1 corresponds approximately to

$$x = \frac{1}{2p+1} \approx \frac{1}{2p}.$$

$$Y_1 = \tfrac{1}{2p}\cdot\tfrac{1}{2p}\cdot\tfrac{2p-1}{2p}\cdot\tfrac{2p}{2p}\cdots\tfrac{p-r}{p+1}\cdots\tfrac{r-1}{p+1} = \tfrac{1}{2p}\,\Gamma\left(1+\tfrac{1}{2p}\right)\Gamma\left(p+1-\tfrac{1}{2p}\right)\cdot\tfrac{1}{\Gamma(p+1)}$$

and

$$\Gamma(p+\tfrac{1}{2}) = (2p)^{p+r-\frac{1}{2}}e^{-2p}\left(1+\tfrac{1}{12p}\right) \sim \sqrt{2\pi}p^p e^{-p+\frac{1}{12p}-\cdots} = \sqrt{2\pi}p^p\,e^{-p+\frac{1}{12p}-\cdots},$$

and

$$(2p+1)^{2p+1} = (2p)^{2p+1}e^{1+\frac{1}{4p}-\cdots}\left(1+\tfrac{1}{2p}\right)$$

so that

$$Y_1 = \frac{\sqrt{2\pi}\,2p^p\,e^{-p+\frac{1}{12p}}}{2^{p+1}p^{p+1}e^{-p}} = \frac{p^{p+1}\sqrt{p}}{e^p}\cdots,$$

so that, p being large, Y_1 is far larger than Y_{p+r}.

[550]

Problem and Hypothetical Theorems in Regard to Two Quadric Surfaces.

[From the Messenger of Mathematics, vol. II. (1873), p. 137.]

Two conics may be circum-and-inscribable to an n-gon; viz. the conics may be such that there exists a singly infinite series of n-gons each inscribed in the first and circumscribed about the second of the conics. In particular they may be circum-and-inscribable to a triangle.

The following problem arises:

Consider two given quadric surfaces and a given line S; to find the planes through S, which cut the surfaces in two conics circum-and-inscribable to a triangle (it is presumed there are two or more such planes).

Let the surfaces be Θ, Θ', and let the line S a tangent to Θ meet Θ in the points A, B; if through S we draw two planes as above, then in the first plane the tangents from A, B to the section of Θ' will meet in a point C of Θ'; and in the second plane the tangents from A, B to the section of Θ' will meet in a point D of Θ'. The points C, D being thus determined the lines AB, AC, BC, AD, BD all touch the surface Θ', and it is presumed that the surfaces Θ, Θ' may be such that CD also touches the surface Θ'; viz. in this case we have a tetrahedron $ABCD$, the summits of which lie in the surface Θ' and the edges touch the surface Θ'; and not only so, but it is further presumed that the surfaces may be such that starting from any point A of Θ and using either any tangent to a properly selected tangent AB of Θ', it shall be possible to complete the figure as above; or, what is the same thing, the surfaces may be such that there exists a doubly or a triply infinite series of tetrahedra, the summits of each lying in Θ and the edges touching Θ'. It is also presumed that the faces of the tetrahedra all touch one and the same quadric surface Θ''.

[551]

Two Smith's Prize Dissertations.

[From the Messenger of Mathematics, vol. II. (1873), pp. 145–148.]

WRITE dissertations on the following subjects:

1. *The theory of interpolation, with a determination of the limits of error in the value of a function obtained by interpolation.*

2. *Determinants.*

The general problem is to find a function of x having given values for given values of x. The problem thus stated is of course indeterminate; in practice, we assume a certain form for the function y, the coefficients of which form are determined by the given conditions, viz. either y is known to be of the form in question, the actual value being then determined as above; or it is assumed that y is approximately equal to a function of the form in question, and the value is then approximately determined in such wise that, for the given values of x, the function y shall have its given values.

The ordinary case is when we have the values of y corresponding to a given value of x, and y is taken to be a function of the form $A + Bx + \cdots + Kx^{n-1}$.

Suppose to fix the ideas $n = 4$, and that p_1, p_2, p_3, p_4 are the values of y corresponding to the values a, b, c, d of x. We may at once write down the expression

$$\begin{aligned} y = p_1 &\frac{(x-b)(x-c)(x-d)}{(a-b)(a-c)(a-d)} \\ + p_2 &\frac{(x-a)(x-c)(x-d)}{(b-a)(b-c)(b-d)} \\ + p_3 &\frac{(x-a)(x-b)(x-d)}{(c-a)(c-b)(c-d)} \\ + p_4 &\frac{(x-a)(x-b)(x-c)}{(d-a)(d-b)(d-c)}, \end{aligned}$$

for clearly this is of the form in question $A + Bx + Cx^2 + Dx^3$, and y becomes $= p_1$ for $x = a$, $= p_2$ for $x = b$, &c. And the like for any value of n. This is known as Lagrange's interpolation formula.

The given values of x may be equidistant, say they are $0, 1, 2, \ldots, n-1$, and the corresponding values of y are y_0, y_1, …, y_{n-1}; then writing down the expression

$$y_x = y_0 + \frac{x}{1}\,\Delta y_0 + \frac{x \cdot x-1}{1 \cdot 2}\,\Delta^2 y_0 + \cdots + \frac{x \cdot x-1 \cdot \ldots \cdot x-n+1}{1 \cdot 2 \cdots n-1}\,\Delta^{n-1} y_0,$$

where, as usual,

$$\Delta y_0 = y_1 - y_0, \quad \Delta^2 y_0 = y_2 - 2y_1 + y_0, \quad \&c.,$$

then for $x = 0$, 1, 2, &c., the values of y are

$$\begin{aligned} &y_0, \\ &y_0 + \Delta y_0, \quad = y_1, \\ &y_0 + 2\Delta y_0 + \Delta^2 y_0, \quad = y_2, \quad \&c., \end{aligned}$$

or the required conditions are satisfied.

As regards the determination of the limits of error, taking a particular case $n = 4$, suppose that we have the values p_1, p_2, p_3, p_4 of y corresponding to the values 0, 1, 2, 3 of x, and that the true value of y is known to be

$$= A + Bx + Cx^2 + Dx^3 + Kx^4,$$

where K is a function of x, which for any value of x within the given values (i.e. from $x = 0$ to $x = 3$) is known to be at least $= P$ and at most $= Q$, i.e. $K = P + tQ$, where t is the ideas. P and Q are each positive. Q being the greater. Hence calculating the interpolation value of $y = Kx^4$ (the last term, the Kx^4 in question), we have

$$\begin{aligned} y = p_1 \frac{1}{6}\,\Delta p_0 + \frac{x \cdot x-1}{1 \cdot 2}\,\Delta^2 p_0 &+ \frac{x \cdot x-1 \cdot x-2}{1 \cdot 2 \cdot 3}\,\Delta^3 p_0 \\ &+Kx^4 \\ &= -\tfrac{1}{6}\,K_0\,x(x-1)(x-2)-3 \\ &\quad +\tfrac{1}{2}\,K_1\,x(x-1)(x-3) \\ &\quad -\tfrac{1}{2}\,K_2\,x(x-2)(x-3) \\ &\quad +\tfrac{1}{6}\,K_3\,x(x-1)(x-3); \end{aligned}$$

viz. this is the true value of y. Hence using the approximate formula as given by the first line, the last four lines give the error, viz. this error is

$$= Kx^4 + K_0\,x(x-1)(x-2)(x-3) + K_1\,x(x-1)(x-2)(x-3) + K_2\,(3Kx^2 - K_3)\,x(x-1)$$
$$-\tfrac{1}{2}\,K_3\,x(x-1)(x-2).$$

But K_0, K_1, K_2, K_3 being each $= P$ and $\leq Q$, this error is

$$\begin{aligned} &> P\{x^4 - 6x^3 + 12x^2 - 8x\} \\ &-Q\{4x^3 - 12x^2 + 12x\}, \end{aligned}$$

and it is

$$< Q\{x^4 - 6x^3 + 12x^2 - 8x\}$$
$$-P\{4x^3 - 12x^2 + 12x\},$$

the difference of these limits being

$$= (Q - P)(x^4 + 22x^3 + 7x^2 + 14x).$$

2. A determinant is a function of n^2 letters; viz. arranging these in the form of a square, the determinant

$$\begin{vmatrix} a_1, & b_1, & c_1, & d_1 \\ a_2, & b_2, & c_2, & d_2 \\ a_3, & b_3, & c_3, & d_3 \\ a_4, & b_4, & c_4, & d_4 \end{vmatrix}$$

is a function linear in regard to each of the n^2 letters, and such that interchanging any two entire lines, or any two entire columns, the sign of the determinant is reversed, its absolute value being unaltered.

The above definition leads to a rule for calculating the actual value of the determinant, which rule may be taken as a definition, viz. the determinant is the sum of $1 \cdot 2 \cdots n$ terms obtained as follows: starting from the term

$$a_1\, b_2\, c_3 \cdots m_n,$$

we permute in every possible way the suffixes $1, 2, 3, \ldots, n$, and give to the term a sign, $\pm$, which is compounded of as many $-$ signs as there are cases in which the arrangement may be obtained by a succession of interchanges of two letters; and then taking for each interchange the sign $-$, we obtain the sign $\pm$ of the term in question. The positive and negative terms are each of them $\frac{1}{2} \cdot 1 \cdot 2 \cdots n$ in number.

To show the execution of the two definitions, it is sufficient to observe that in the second definition, attending for instance only to the first and second columns, in any term $\Delta a_i b_j$, $\Delta a_j b_i$, &c., always correspond either terms $= \Delta a_i b_j$, &c., so that taking the pairs together, there are $\Delta(a_i b_j - a_j b_i)$, $\Delta(a_i b_k - a_k b_i)$, &c., terms which change their sign, but remain unaltered as to their absolute values by the interchange of the first and second columns.

Among the fundamental properties of determinants are as follows:

The properties are the same as regards lines and columns.

A determinant vanishes if any line vanishes (that is, if each term of the line is $= 0$).

A determinant vanishes if two lines are identical.

A determinant is a linear function of its lines.

Wherever:

Determinant having a line $A\ \alpha + a$ times the determinant having the line A ($\alpha\beta$ is here used to denote the line each term of which is α times the corresponding term of the line A).

Determinant having a line $A + a$ = determinant with the A + determinant with the line A'.

It follows that, if any line of a determinant is the sum of the other lines, each multiplied by an arbitrary coefficient, or, what is the same thing, if we can with any of the lines, each multiplied by an arbitrary coefficient, compose a line 0, then the determinant is $= 0$.

The same principle leads to a theorem for the product of two determinants of the same order n, viz. it is found that the product is a determinant of the same order n, each term thereof being a sum of the products of the terms of a line of one of the factors into the corresponding terms of a line of the other factor. Starting with this expression of the product, we decompose it into a series of determinants each of which is either $= 0$ or it is a product of a single term of the one factor into the other factor; and the sum of all these products is equal to the product of the two factors.

The unit determinant is a theorem in which we have a quantities x, y, $\ldots$ connected by the linear equations

$$ax + by + cz + \cdots = 0,$$

$$a_1x + b_1y + c_1z + \cdots = 0,$$

$$a_2x + b_2y + c_2z + \cdots = 0;$$

and so, if we have a linear equations

$$ax + by + cz + \cdots = u,$$

then each of the quantities x, y, z, $\ldots$ is given as the quotient of two determinants, the denominator being in each case

$$\begin{vmatrix} a, & b, & c, & \cdots \\ a_1, & b_1, & c_1, & \cdots \\ a_2, & b_2, & c_2, & \cdots \\ \vdots & \vdots & \vdots & \ddots \end{vmatrix}$$

and the numerators being (save as to their signs) that for x

$$\begin{vmatrix} u, & b, & c, & \cdots \\ 0, & b_1, & c_1, & \cdots \\ 0, & b_2, & c_2, & \cdots \\ \vdots & \vdots & \vdots & \ddots \end{vmatrix}$$

and the like for y, z, $\ldots$.

A determinant remains unaltered when the lines and columns are interchanged, the dexter diagonal $\backslash$ remaining unaltered.

A determinant

$$\begin{vmatrix} a_1, & b_1, & c_1, & d_1, & \cdots \\ a_2, & b_2, & c_2, & d_2, & \cdots \\ a_3, & b_3, & c_3, & d_3, & \cdots \\ a_4, & b_4, & c_4, & d_4, & \cdots \end{vmatrix}$$

is a sum of products of complementary determinants, viz.

$$\Sigma \pm \begin{vmatrix} a_i, & b_j \end{vmatrix} \begin{vmatrix} c_k, & d_l, & \cdots \end{vmatrix},$$

or, say of products of complementary minors.

In particular, it is a sum

$$\Sigma \pm a_i \begin{vmatrix} b_j, & c_k, & d_l, & \cdots \end{vmatrix},$$

of products of first minors into single terms or $(n-1)^{\text{th}}$ minors.

This last theorem affords a convenient rule for the development of a determinant.

[552]

On a Differential Formula Connected with the Theory of Confocal Conics.

[From the Messenger of Mathematics, vol. II. *(1873), pp. 157, 158.]*

THE following transformations present themselves in connexion with the theory of confocal conics.

The coordinates x, y of a point are considered as functions of the parameters h, k where

$$\frac{x^2}{a+h}+\frac{y^2}{b+h}=1,$$

$$\frac{x^2}{a+k}+\frac{y^2}{b+k}=1;$$

and then assuming $\sqrt{a+h}=s$, $\sqrt{a+k}=t$, ($i=\sqrt{-1}$ as usual) and writing $c=a-b$, we have

$$\begin{aligned}H&=(s+i\sqrt{c-s^2})(t+i\sqrt{c-t^2}),\\ K&=(s+i\sqrt{c-s^2})(t-i\sqrt{c-t^2}),\end{aligned}$$

whence if

$$H=(s+h)(s+k),\quad K=(s+h)(s-k),$$

we have

$$\begin{aligned}H&=\tfrac{1}{4}\{\sqrt{c}(s+t)+i\sqrt{(c-s^2)(c-t^2)}\}^2,\\ K&=\tfrac{1}{4}\{\sqrt{c}(s-t)+i\sqrt{(c-s^2)(c-t^2)}\}^2,\end{aligned}$$

or, say

$$\begin{aligned}\sqrt{H}&=\tfrac{1}{2}\{\sqrt{c}(s+t)+i\sqrt{(c-s^2)(c-t^2)}\},\\ \sqrt{K}&=\tfrac{1}{2}\{\sqrt{c}(s-t)+i\sqrt{(c-s^2)(c-t^2)}\}.\end{aligned}$$

and thence

$$\begin{aligned}
k + \tfrac{1}{2}(s+h) + \sqrt{H} &= \tfrac{1}{2}\{i\sqrt{(c-s^2)(c-t^2)} - i + a\sqrt{(s-t)}\},\\
k + \tfrac{1}{2}(s+h) + \sqrt{K} &= \tfrac{1}{2}\{i\sqrt{(c-s^2)(c-t^2)} + i + a\sqrt{(s-t)}\},\\
&= -[s + \tfrac{i}{2}\sqrt{(c-s^2)}] \cdot [t + \tfrac{i}{2}\sqrt{(c-t^2)}].
\end{aligned}$$

These also follow from the known differential formula

$$4(ds^2 + dy^2) = -kh\left(\frac{ds^2}{H} - \frac{dt^2}{K}\right),$$

that is,

$$\frac{4dtdy}{\sqrt{(c-s^2)(c-t^2)}} = \frac{ds}{\sqrt{H}}\,\frac{dt}{\sqrt{K}},$$

implying

$$\frac{2tdy}{\sqrt{(c-s^2)}}\,\frac{di}{\sqrt{(c-t^2)}} = \frac{ds}{\sqrt{H}} - \frac{dt}{\sqrt{K}},$$

$$\frac{2dy}{\sqrt{(c-t^2)}} + \frac{ds}{\sqrt{H}}\,\frac{dt}{\sqrt{K}} = \frac{ds}{\sqrt{H}} + \frac{dt}{\sqrt{K}},$$

where a is a constant. The foregoing integral formula give at once

$$\frac{ds}{\sqrt{H}} - \frac{dt}{\sqrt{K}} = \frac{dy}{\sqrt{(c-t^2)}},$$

$$\frac{ds}{\sqrt{H}} + \frac{dt}{\sqrt{K}} = \frac{ds}{\sqrt{(c-s^2)}},$$

and substituting these values we find $a = 1$, and the differential formula are then satisfied.

We thence have

$$\text{const.} = \sqrt{(a+h)(b+h)} + \sqrt{(a+k)(b+k)},$$

as the integral of the differential equation

$$\frac{dh}{\sqrt{H}} + \frac{dk}{\sqrt{K}} = 0,$$

[553]

Two Smith's Prize Dissertations.

[From the Messenger of Mathematics, vol. II. (1873), pp. 161–166.]

WRITE dissertations:

1. *On the condition of the similarity of two physical systems.*

2. *On orthogonal surfaces.*

1. Considering two particles m, m_1, describing similar paths similar parts (which for convenience may be taken to be similarly situate in regard to two axes of rectangular axes respectively); viz. this means that the times t, t' of passage through corresponding arcs s, s' are proportional. The ratios $\frac{s}{s'}$, $\frac{t}{t'}$ are then each of them constant; and this must also be the case with the ratio $\frac{v}{v'}$, of the velocities v, v' at corresponding points; since it is clear that we must have $\frac{v}{v'} = \frac{s/s'}{t/t'}$.

Now in order that the two particles may move as above under the action of any forces upon the two particles respectively, it is clearly necessary that the forces F, F' at corresponding points shall act in the same direction, and be in a constant ratio of magnitudes. To obtain this ratio, imagine the two particles, m, m' moving as above, in the corresponding infinitesimal elements of times τ, τ', with the velocities u, u' through the infinitesimal arcs σ, σ' respectively; $\frac{u}{u'} = \frac{s/s'}{\sigma/\sigma'} = \frac{u}{u'}$; and the deflections from the tangent will be $\frac{1}{2}\,\frac{F}{m}\,\tau^2$, $\frac{1}{2}\,\frac{F'}{m'}\,\tau'^2$ respectively, and these must be in the ratio of the corresponding arcs σ, σ', viz. we must have

$$\frac{F\tau^2}{m} : \frac{F'\tau'^2}{m'} = \sigma : \sigma'.$$

or, what is the same thing,

$$\frac{Ft}{m} = \frac{F't'}{m'} \cdot \frac{\sigma}{\sigma'} : \frac{\sigma'}{s'}$$

that is,

$$\frac{F}{F'} = \frac{m}{m'} \cdot \frac{s}{s'} \cdot \left(\frac{t}{t'}\right)^2,$$

and this relation subsisting, we have at corresponding points of the paths of the two particles of particles moving under the like geometrical conditions, and we have above at the conclusion, given two physical constituted systems, which act upon any masses in a given magnitude-ratio; the component-particles being in a given ratio: $\dfrac{m}{m'}$ (the same for each pair of component particles); then if the particles of the two systems respectively are to move in similar paths of the same magnitude-ratio $\dfrac{s}{s'}$ the times of describing corresponding arcs being in a given constant ratio $\dfrac{t}{t'}$ (this implying as above that the ratio of the velocities at corresponding points is $\dfrac{v}{v'} = \dfrac{s/s'}{t/t'}$), it is necessary that the forces on corresponding particles in corresponding positions shall act in the same direction, and shall be in the constant magnitude-ratio

$$\frac{F}{F'} = \frac{m}{m'} \cdot \frac{s}{s'} \cdot \left(\frac{t}{t'}\right)^2,$$

and this being so, the motion of the two systems will in fact be similar as above explained.

Taking $\dfrac{m}{m'} = \alpha$ for the mass-ratio, $\dfrac{s}{s'} = a$ for the length-ratio, and $\dfrac{t}{t'} = \tau$ for the time-ratio; also $\dfrac{F}{F'} = \phi$ for the force-ratio, the condition determining the force-ratio ϕ is then

$$\phi = \frac{\alpha a}{\tau^2}.$$

It is to be observed, that if the forces are entirely internal, and proportional to homogeneous functions of the same order, say $-r$, of the coordinates of all or any of the particles; e.g. if they are central forces varying as the inverse nth power of the distance; then the condition as to the action of the forces in the two systems respectively can always be satisfied by giving a proper constant value to the ratio of the absolute forces (or forces at unity of distance), thus, if in the first system we have two particles m_1, m_2, attracting or repelling each other with a force $\dfrac{km_1m_2}{r^n}$, and if in the second system the force is $\dfrac{k'm_1'm_2'}{r'^n}$, then the condition as to the direction of the forces at corresponding positions is satisfied (*qua facts*); and the condition as to magnitude is

$$\frac{km_1m_2/r^n}{k'm_1'm_2'/r'^n} : \frac{m_1}{m_1'} \cdot \frac{r}{r'} \cdot \frac{1}{r^2} = 1 = \phi,$$

that is,

$$\frac{k}{k'} = \frac{m_1'}{m_1} \cdot \frac{r'^{n-1}}{r^{n-1}} \cdot \frac{t}{t'} \cdot \phi,$$

or, say

$$\frac{k}{k'} = \left(\frac{m_1'}{m_1}\right)^{n-1} \left(\frac{r'}{r}\right)^{n-2} \left(\frac{t}{t'}\right)^2.$$

In the case $n = 2$, the constant therein (applying however only to the case of two elliptic orbits of the same eccentricity) agrees with Kepler's third law, or say with the theorem

$$T = \frac{2\pi a^{3/2}}{\sqrt{(kM)}},$$

that is,

$$T = \frac{2\pi}{k} a^{3/2}\ \sqrt{(M)},$$

where observe that the μ, or mass in the sense of the formula, is the km, or attractive force on a unit of mass, of the theorem as above written down.

2. In a family of surfaces $F(x, y, z, p) = 0$, containing a single variable parameter p, there is through any given point of space, a surface or surfaces of the family; or (if more than one, confining the attention to one of these surfaces) we may say that there is, through any given point of space, a surface of the family.

Considering now two other families of surfaces, there will be through any given point of space, three surfaces, one of each family; and if the every given point of space (whatever) there intersect each other at right angles, we have a system of three orthogonal surfaces.

Supposing the equations of the three families to be

$$\begin{aligned} F(x, y, z, p) &= 0, \\ \Phi(x, y, z, q) &= 0, \\ \Psi(x, y, z, r) &= 0, \end{aligned}$$

then the requisite conditions are

$$\frac{dF}{dx}\frac{d\Phi}{dx}+\frac{dF}{dy}\frac{d\Phi}{dy}+\frac{dF}{dz}\frac{d\Phi}{dz}=0,$$

$$\frac{dF}{dx}\frac{d\Psi}{dx}+\cdots=0,$$

$$\frac{d\Phi}{dx}\frac{d\Psi}{dx}+\cdots=0,$$

viz. these equations must be satisfied not in general identically, but in virtue of the given equations $F=0$, $\Phi=0$, $\Psi=0$.

Or, what is more convenient, we may take the equations of the three families to be

$$p=f(x,y,z)=0,\quad q=\phi(x,y,z)=0,\quad r=\psi(x,y,z)=0,$$

and write the conditions in the form

$$\frac{dp}{dx}\frac{dq}{dx}+\frac{dp}{dy}\frac{dq}{dy}+\frac{dp}{dz}\frac{dq}{dz}=0,$$

$$\frac{dp}{dx}\frac{dr}{dx}+\cdots=0,$$

$$\frac{dq}{dx}\frac{dr}{dx}+\cdots=0,$$

where of course p, q, r stand for their given functions $f(x,y,z)$, $\phi(x,y,z)$, $\psi(x,y,z)$; the equations in this form contain only (x,y,z), and not the parameters p, q, r; so that, if satisfied at all, they must be satisfied identically; and the required conditions therefore are that the last-mentioned system of equations shall be satisfied identically by the values p, q, r considered as given functions of (x,y,z).

The last-mentioned conditions lead to the theorem known as Dupin's; viz. it follows from them that the surfaces intersect along their curves of curvature; or more definitely, each surface of one family is intersected by the surfaces of the other two families in its two sets of curves of curvature.

To indicate the geometrical ground of the theorem, consider on a surface of one family a point P, and at this point the normal meeting the consecutive surface in P'; the surfaces through P of the other two families respectively will pass through P', and meet the given surface in two curves PA, PB (= PA', PB' represent infinitesimal arcs on these two curves respectively, the angle at P being a right angle.

Drawing at A, B normals to the given surface to meet the consecutive surface in the points A', B' respectively, the same two normals will meet the consecutive surface in the points P, P', respectively; and (the system being orthogonal), we must have the angle at P' a right angle. This imposes a condition upon the direction (in the tangent plane at P) of the orthogonal directions PA, PB; viz. it is found that these must be such that the two normals PP', AA' intersect, or, what is the same thing, the bisecting directions PA and PB must be the directions of the two curves of curvature.

Observe that PP', AA' intersecting each other, the four points P, P', A, A' are in the same plane, that is, PA, $P'A'$ intersect, from these being the normals at P, P' respectively of the surface through P or one of the other two families; and similarly PP', BB' intersecting each other, this, the four points P, P', B, B' are in the same plane, that is, PB, $P'B'$ intersect, these being the normals through P, P' to the surface through P of the third family.

Through any of the points P, A, A', P', viz. in the form

$$PA' \cdot PA + PP' \cdot AA' = PA \cdot PA' + PP' \cdot AA',$$

on the tangent plane at P of the orthogonal directions PA, PB; viz. it is found that these must be such that the two normals PP', BB' intersect, and is the same thing, the bisecting directions PA and PB must be the directions of the two curves of curvature.

Observe that PP', BB' intersecting each other, the four points P, P', B, B' are in the same plane, that is, PB, $P'B'$ intersect from these being the normals at P, P' respectively of the surface through P of one of the other two families; and similarly the points C, D, E, F are coplanar (the cycle for one surface, and the surface for the orthogonal PR, $P'R'$), then this P, P', R, R' are also coplanar, (PR being a right angle, also as is a a normal there are the directions of the curves of curvature at the point PR, the chord drawn in the directions of the curves of curvature on the consecutive surface. But in the orthogonal system they must be so; and this imposes upon the infinitesimal normal distance p, a condition; viz. it is found that p considered as a function of (x, y, z) must satisfy a certain partial differential equation of the second order.

It hence appears that no one of the three families of surfaces can be assumed at pleasure; for taking the equation of a family to be $p = f(x, y, z) = 0$, there is being the value of the parameter for the given surface of the foregoing investigation, and $p + dp$ the value of the parameter for the consecutive surface, the normal distance at the point (x, y, z) between the two surfaces is

$$= dp = \sqrt{\left[\left(\frac{dp}{dx}\right)^2 + \left(\frac{dp}{dy}\right)^2 + \left(\frac{dp}{dz}\right)^2\right]},$$

viz. dp is here a constant; and we have

$$1 = \sqrt{\left[\left(\frac{dp}{dx}\right)^2 + \left(\frac{dp}{dy}\right)^2 + \left(\frac{dp}{dz}\right)^2\right]};$$

satisfying the foregoing partial differential equation; or, what is the same thing, p considered as a function of x, y, z must satisfy a certain partial differential equation of the third order; viz. this is the condition to be satisfied in order that a family of surfaces $p = f(x, y, z) = 0$ may belong to an orthogonal system.

[554]

An Elliptic-Transcendent Identity.

[From the Messenger of Mathematics, vol. II. (1873), p. 178.]

THE following is a singular identity:

$$
\begin{aligned}
&(1+q)(1+q^2)(1+q^4)(1+q^7)(1+q^9)\cdots \\
&-(1-q)(1-q^2)(1-q^4)(1-q^7)(1-q^9)\cdots \\
&=2q(1+q^2)(1+q^4)(1+q^6)(1+q^8)(1+q^7)(1+q^9)\cdots\,,
\end{aligned}
$$

where in each of the three terms every factor has the exponent 1 or 2 according as the exponent of q is not, or is, divisible by 7.

[555]

Notices of Communications to the British Association for the Advancement of Science.

[From the Reports of the British Association for the Advancement of Science, 1865 to 1872. Notices and Abstracts of Miscellaneous Communications to the Sections.]

1. On the Problem of the in-and-circumscribed Triangle. Report, 1870, p. 10.

I have recently accomplished the solution of this problem, which I spoke of at the Meeting in 1864. The problem is as follows: required the number of the triangles the angles of which are situate in a given curve or curves, and the sides of which touch a given curve or curves. There are in all 32 cases [see 514] of the problem, according as the curves which contain the angles and are touched by the sides are distinct curves, or are two of all the same, that is two of them all distinct, &c. when the curves are all of them distinct, the number of triangles is here $= 2mn(2nD\bar{a}D'\bar{a}^4$, where a, a' are the orders of the curves containing the angles (or, say, of the *angle-curves* respectively); and β, γ' are the classes of the curves touched by the sides (or, say, of the *side-curves* respectively); and these have other classes referring to a like formula, viz. the number of triangles is here $= 2(B-1)(D-2)$ &c.; where, if B, γ are the class and order of the same curve $B = D - F$. The simplest case is when the curves are all of them one and the same curve, say $a = a' = a'' = B = D = F$, the number of triangles is here equal to

$$\begin{aligned}
&+ a^2 \,(\qquad\qquad\qquad\qquad +1),\\
&+ aB^2(\quad +2b - 18b + 12c - 8d)\\
&+ aB'(\quad -18b^2 + 102bc - 432bd + 270cd)\\
&+ aB''(-16b^2 + 252b^2c - 622bcd + 624bd^2 - 432cd)\\
&+ B''(\qquad\qquad\quad -8)\\
&+ \{B^4(\quad +18b - 192c)\}\\
&+ \{B^3(\quad -38b + 1152c)\}.
\end{aligned}$$

where a is the order, A the class of the curve; α is the number, three times the class + the number of cusps, or (what is the same thing) three times the order + the number of inflexions.

2. On a Correspondence of Points and Lines in Space. Report, 1870, p. 10.

NINE points in a plane may be the intersection of two (and therefore of an infinite series of) cubic curves; say, that the nine points are an "encead": so, and similarly nine lines through a point may be the intersection of two (and therefore of an infinite series of) cubic cones; say, the nine lines are an encead. Now, imagine (in space) any 8 given points; taking a variable point P, and joining this with the 8 points, we have through P 8 lines and there is through P a ninth line completing the encead; this is and to be the corresponding line of P. We have thus to any point P a single corresponding line through the point P; this is the correspondence referred to in the heading, and which I would suggest as an interesting subject of investigation to geometers. Observe that, considering the whole system of points in space, the corresponding lines are a triple system of lines, not the whole system of lines in space. It is thus, not any line whatever, but only a line of the triple system, which has on it a corresponding point. But as to this same explanation is necessary; for starting with an arbitrary line, and taking upon it a point P, it would seem that P might be so determined that the given line and the lines from P to the eight points should form an encead,—that is, that the arbitrary line would have upon it a corresponding point or points.

The question of the foregoing species of correspondence was suggested to me by the consideration of a system of 10 points, such that joining any one whatever of them with the remaining nine points, the nine lines from this obtained form an encead; or, say, that each of the 10 points is the "enneadic centre" of the remaining nine. I have been led to such a system of 10 points by my researches upon Quartic Surfaces; but I do not as yet understand the theory.

3. On the Number of Covariants of a Binary Quantic. Report, 1871, pp. 9, 10.

THE author remarked [see 462] that it had been shown by Prof. Gordan that the number of the covariants of a binary quantic of any order was finite, and, in particular, that the numbers for the quintic and the sextic were 23 and 26 respectively. But the demonstration is a very complicated one, and it can scarcely be doubted that a more simple demonstration will be found. The question in its most simple form is as follows: viz. instead of the covariants we substitute their leading coefficients, each of which is a "seminvariant" satisfying a certain partial differential equation; say, the quantic is $(a, b, c, \ldots, k)(x, y)^n$, then the differential equation is $(a\partial_b+2b\partial_c+\cdots+nj\partial_k)u = 0$, which ∂ quasi equation with $n+1$ variables admits of n independent solutions: for instance, if $n = 3$, the equation is $(a\partial_b + 2b\partial_c + 3c\partial_d)u = 0$, and the solutions are a, $ac - b^2$,